

Seminar: Algebraic Curves

Talk 11: Further applications of Bézout's theorem

Nico Fischer

Given a curve F in $\mathbb{P}^2(K)$ and a point P such that $\mathcal{O}_{F,P}$ is an integral domain, one can, if G is a curve sharing no component with F through P , describe an alternative way of calculating $\mu_P(F, G)$ using valuations from DVRs in $\mathfrak{X}(F)$ that lie over P .

Using this description of $\mu_P(F, G)$, further calculations lead to:

Theorem (7.4). *Let F, G be two curves in $\mathbb{P}^2(K)$ and P any point. Then*

$$\mu_P(F, G) \geq m_P(G)m_P(F).$$

Equality holds if and only if F and G share no tangent lines at P .

In particular, combining this with Bézout's theorem yields:

Corollary (7.5). *If F and G have no components in common, then*

$$\deg(F) \cdot \deg(G) \geq \sum_{P \in \mathbb{P}^2(K)} m_P(F) \cdot m_P(G).$$

Taking these results as granted, we will investigate how many singularities an irreducible curve of degree d can have.

Definition (7.12). *Let L be a set of curves of degree d in $\mathbb{P}^2(K)$. If there is a subspace V of the K -vector space of homogeneous degree d polynomials in $K[X_0, X_1, X_2]$ such that L consists of all curves defined by polynomials in $V \setminus \{0\}$, then L is called a linear system of degree d . We furthermore define $\dim L := \dim V - 1$.*

If we are given points $P_1, \dots, P_s$ and integers $m_1, \dots, m_s \geq 1$, we can consider linear systems of curves passing through each P_i with multiplicity at least m_i . Counting degrees of freedom yields:

Theorem (7.14). *The curves F of degree d with $m_{P_i}(F) \geq m_i$ for $i \in \{1, \dots, s\}$ form a linear system L with*

$$\dim L \geq \binom{d+2}{d} - \sum_{i=1}^s \binom{m_i+1}{2} - 1.$$

To get an upper bound for the number of singular points of an irreducible curve, we first consider:

Theorem (7.9). *Let F be an irreducible curve in $\mathbb{P}^2(K)$ of degree d , and let $P_1, \dots, P_s$ be its singular points. Then*

$$\sum_{i=1}^s m_{P_i}(F)(m_{P_i}(F) - 1) \leq d(d-1).$$

Combining this result with Theorem 7.14, and Theorem 7.5, one obtains:

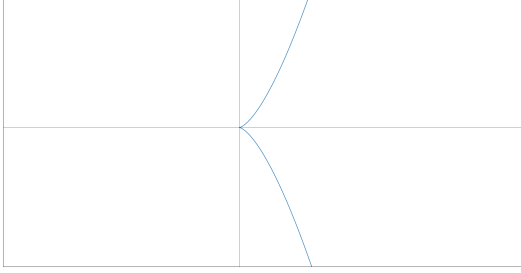
Theorem (7.15). *Let F be an irreducible curve of degree d , and let $P_1, \dots, P_s$ be its singular points. Then*

$$\binom{d-1}{2} \geq \sum_{i=1}^s \binom{m_{P_i}(F)}{2}.$$

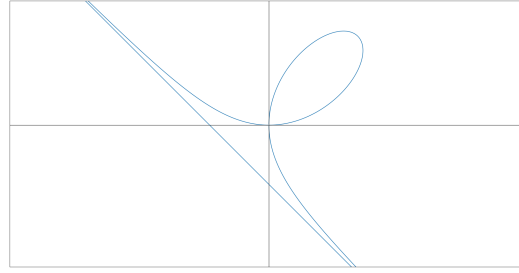
As a corollary, we get the promised bound on the number of singular points:

Corollary (7.16). *An irreducible curve of degree d has at most $\binom{d-1}{2}$ singularities.*

Example. *Consider an irreducible cubic. It has up to one singularity of multiplicity two. The singular point either has two distinct tangents or a single double tangent. Both situations are illustrated below.*



Neil's Semicubical Parabola



Folium of Descartes

The initially mentioned results connecting valuations and intersection multiplicity can be used to give a sufficient condition for the intersection scheme of two curves to be a subscheme of a third.

Theorem (7.18). *Let F_1, F_2, G be curves in $\mathbb{P}^2(K)$ and suppose one of the following conditions is satisfied for a point $P \in \mathcal{V}_+(F_1) \cap \mathcal{V}_+(F_2)$:*

- *P is a simple point of F_1 and $\mu_P(F_1, G) \geq \mu_P(F_1, F_2)$.*
- *F_1 and F_2 have no tangents in common at P and $m_P(G) \geq m_P(F_1) + m_P(F_2) - 1$.*

Then

$$\dim_K \mathcal{O}_{F_1 \cap F_2 \cap G, P} = \mu_P(F_1, F_2).$$

And lastly, we get the corollary:

Corollary (7.19). *Let F_1 and F_2 be curves with no component in common, and G another curve. For each $P \in \mathcal{V}_+(F_1) \cap \mathcal{V}_+(F_2)$, suppose one of the following conditions is satisfied:*

- *F_1 is smooth at P and $\mu_P(F_1, G) \geq \mu_P(F_1, F_2)$.*
- *F_1, F_2 have no tangent line in common at P and $m_P(G) \geq m_P(F_1) + m_P(F_2) - 1$.*

Then $F_1 \cap F_2$ is a subscheme of G .

References

- [1] E. Kunz. *Introduction to plane algebraic curves*. Birkhauser, 2005. Translated from the 1991 German edition by Richard G. Belshoff, Information for the present document primarily drawn from Section 7.